Is the documented temperature compensation good for the prediction system?

Study 2 — the legacy instrument era, station 02, calendar year 2021
Medieval walls of Gubbio, static structural health monitoring

Notebook executed 11 August 2026 · Data window 1 January – 31 December 2021 · Hourly analysis grid

Summary

This report asks the same questions as its companion study on the current instrument era, on different data. The correction under test is the fixed thermal correction that `00_Sensor_Preprocessing` applies before any other stage sees the signal:

$$I_{\mathrm{comp}}(t) \;=\; I(t) \;-\; \left(T_{\mathrm{air}}(t) - T_{\mathrm{air}}(t_0)\right) \cdot c \cdot 1000, \qquad c = 0.005. \tag{1}$$

Where the current-era study had four environmental channels but only half an annual cycle, this one has a *full calendar year* and only two channels. That trade is the reason to run it: the seasonal response is observable here and was not observable there.

Verdict on the correction: it fails all three criteria, on independent data

Criterion	Measured	Result
Thermal dependence is reduced	$\|r\|$: 0.238 → 0.738	fail
Coefficient is near the variance-minimising value	5.00 vs 1.00 mdeg/°C	fail
Forecast improves under a controlled contrast	penalty +0.7 % to +47.4 %	fail

Three for three, matching the current-era study on a different instrument and a different set of regressors.

The finding only a full year could produce: the structure *leads* the air

Over 2021 the inclination correlates with air temperature at 0.238 in **levels** but 0.866 in **first differences**. The reason is a phase difference, and its direction is the important part. A 30-day mean of the inclination peaks on **7 June**; a 30-day mean of air temperature peaks on **5 August**. The structure reaches its seasonal maximum **59 days before** the air does, and a constrained daily cross-correlation independently puts the lead at 89 days.

No thermal lag can represent this. A causal filter can only delay a driver, never advance it, and the scan confirms it: every time constant from 1 to 168 hours makes the annual fit monotonically *worse*, from $R^2 = 0.056$ at $\tau = 0$ down to 0.000. The annual movement of this wall is not a delayed response to air temperature at all.

It is, however, what the physical hypothesis predicts. If the forcing is solar gain on an exposed vertical face, its annual maximum arrives well before the annual maximum of air temperature, because irradiance on a vertical surface peaks when the sun is lower than at the solstice. Air temperature is a good proxy for the forcing from hour to hour — at the diurnal scale it explains 91.5 % of the variance, instantaneously — and a poor one from month to month.

Recommendation. Remove the fixed compensation from Notebook 00. Beyond that, the two timescales need different treatment: the daily response can be carried by instantaneous air temperature, while the seasonal component must be left to the decomposition's own annual term or driven by a radiation-like variable, because no coefficient and no lag on air temperature can reproduce a lead.

Contents

1 What differs from the current-era study

Table 1: The two studies, side by side.

	Current era (Study 1)	Legacy era (this study)
Period	from 2025-02-21	before 2025-02-21
Environmental channels	`tair`, `rh`, `sr`, `twall`	`tair`, `rh` only
Usable span	197 days in two blocks	a full calendar year
Seasonal response	not observable	observable
Gap structure	two multi-month outages	84 gaps, none over 24 hours
Instrument	installed 2025-02-21	legacy block `b2`

The legacy record shares no baseline with the current one: the 2025 installation was a physical reinstallation, so the absolute levels are unrelated. That is not a limitation here, because every quantity this report relies on — correlations, slopes, variances, forecast errors — is invariant to the arbitrary offset.

Sampling. As in the companion study, the record is aggregated from its native 20-minute rate to one hour immediately after parsing, matching the hourly rate of every external proxy the pipeline aligns against. An hour is kept only if at least two of its three raw samples survived cleaning.

Shared tests. The acceptance tests, coefficient sweep, slope fitting, lag and inertia analysis and forecasting experiment are *imported* from the sibling study rather than reimplemented: `lc_lib.py` adds only the legacy loader, the year-completeness scan, the decomposition and the imputation. Any difference in results between the two reports is therefore a difference in the data, not in the method.

2 Choosing the year

Whole-day file counts mislead here, because a day can exist as a file while station 02's own block is written as zeros throughout it. The scan measures what the study actually needs: hourly intervals in which the inclinometer, air temperature and humidity are *all* present.

Only three years required measuring. From the archive audit, 2018 covers 159 days (acquisition began in July), 2022, 2023 and 2024 miss 212, 171 and 82 whole days respectively, and the legacy record ends 51 days into 2025.

Table 2: Usable completeness by year, station 02, on the hourly grid.

Year	Hours	Complete	Complete [%]	Longest run [days]	Verdict
2021	8 737	8 516	**97.47**	66.6	**selected**
2019	8 737	7 505	85.90	34.0	
2020	8 761	7 270	82.98	43.8	

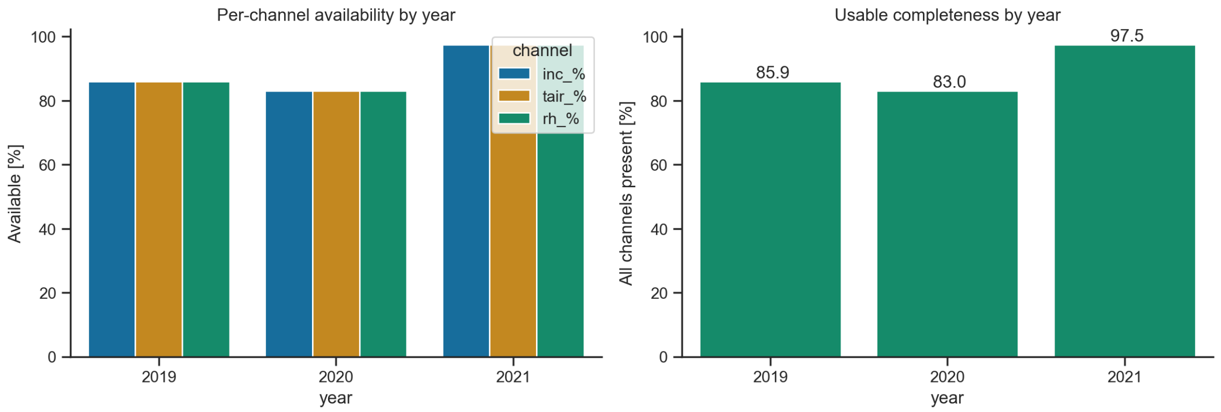

Figure 1: Left, per-channel availability by year — the three channels fail and recover together, so the losses are acquisition outages rather than individual sensor faults. Right, usable completeness, the quantity that decides the selection.

The equal bar heights in the left panel are informative in themselves: the three channels are available in exactly the same proportion in every year, which is the signature of whole-station dropouts rather than channel-specific failure. Nothing needs to be modelled channel by channel.

2021 is not merely the most complete year, it is the best-shaped one, and that matters more than the headline percentage.

Table 3: Gap profile of the inclinometer channel in 2021, on the hourly grid. The other two channels are identical.

Gap length	Gaps	Hours lost	Share of year
Single hour	44	44	0.5 %
Up to 6 h	33	108	1.2 %
6–24 h	7	69	0.8 %
Total	**84**	**221**	**2.5 %**

No gap in 2021 exceeds 24 hours, and the longest unbroken run is 66.6 days. That is what makes the NeuralProphet imputation in Section 4 genuine interpolation between known values rather than extrapolation from a seasonal model.

3 The year as recorded

Table 4: Channel coverage over 2021 (8 737 hourly intervals).

Channel	Present	Available	Min	Max	Mean	Std
`batt`	8 516	97.5 %	3.400	4.690	3.891	0.297
`tair`	8 516	97.5 %	−4.63	38.48	13.710	8.818
`rh`	8 516	97.5 %	19.65	99.94	71.908	22.111
`inc`	8 516	97.5 %	−338.67	−205.42	−272.46	33.891

The inclination standard deviation over the full year, 33.9 mdeg, is more than twice the 14.6 mdeg measured over the current era's Window A. The difference is not instrumental — it is the annual cycle, which Window A only samples in part. The spike screen removed nothing from this year: with a robust σ of 43.3 mdeg the threshold sits at 346 mdeg, and no hourly value approaches it.

Figure 2: The 2021 record: inclination, air temperature, relative humidity and battery voltage on a common time axis.

Figure 2 shows the shape the rest of the report analyses, and the central fact is already visible. The inclination rises through spring, reaches its highest values in late May and June, and then turns down — while air temperature is still climbing towards its August maximum. The two series do not peak together. Section 5 measures the difference.

4 Decomposition and imputation

4.1 NeuralProphet decomposition of every series

Each series is fitted *without* autoregression, so that every component is a function of time alone. That restriction is deliberate: it keeps the components interpretable, and it is what allows the same fit to be used for imputation, since a model with autoregressive terms cannot produce a value inside a gap.

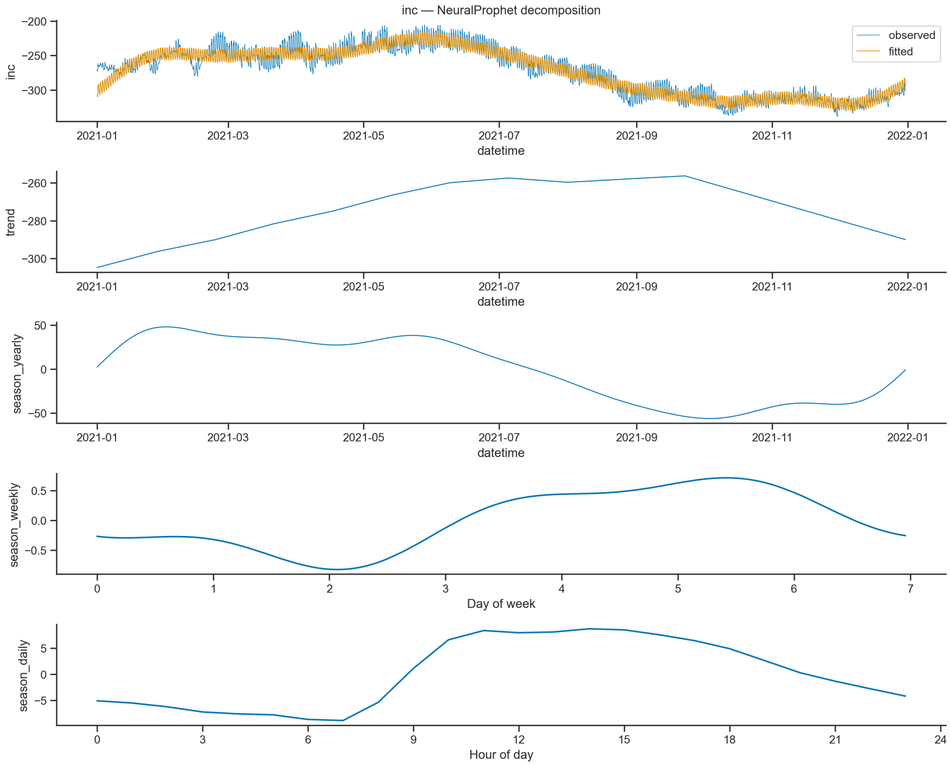

Figure 3: Inclination, decomposed. From the top: observed and fitted, trend, annual term, weekly term, daily term.

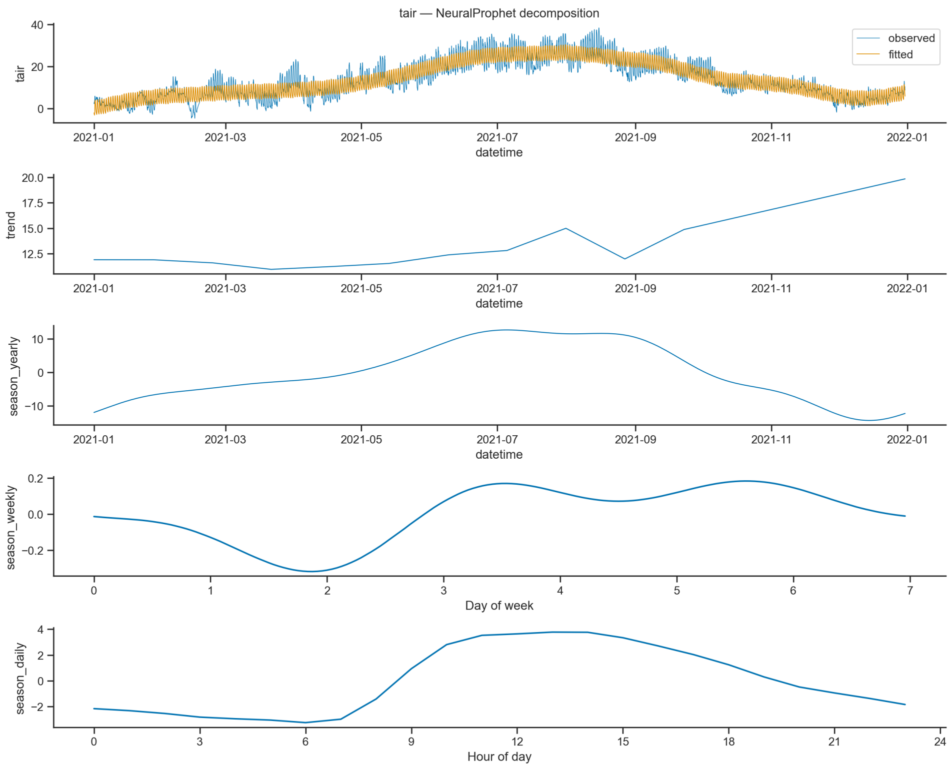

Figure 4: Air temperature, decomposed on the same basis.

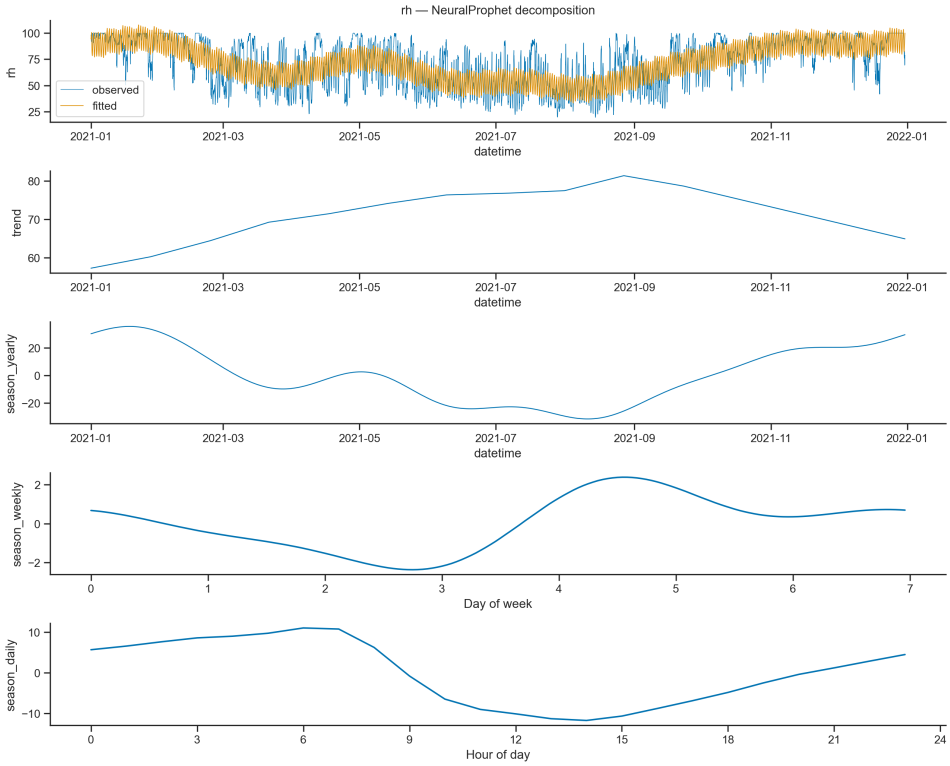

Figure 5: Relative humidity, decomposed on the same basis.

Table 5: Decomposition summary. Component standard deviations are in the native units of each series.

Series	Var. obs.	Var. resid.	Explained	Trend	Yearly	Weekly	Daily
`inc`	1148.583	87.735	92.4 %	14.035	**36.131**	0.487	6.438
`tair`	77.750	10.399	86.6 %	2.551	8.398	0.155	2.554
`rh`	488.902	208.520	57.3 %	6.318	20.124	1.358	7.756

Three things follow.

The inclination is an annual signal first and a daily signal second. Its yearly term has a standard deviation of 36.1 mdeg against 6.4 mdeg for the daily term — a factor of five and a half. Trend and seasonality together explain 92.4 % of its variance. For a structure whose monitoring question is slow drift, that ratio is the most important number in the decomposition: the annual environmental cycle is the dominant thing any anomaly must be detected against.

The weekly term is negligible. 0.487 mdeg for the inclination and 0.155 °C for air temperature, against daily terms thirteen to sixteen times larger. A structure with no human loading cycle and a rural weather station should have no weekly signature, and they do not. This is a free check that the decomposition is not manufacturing components.

Humidity is the least deterministic channel. Only 57.3 % of its variance is explained by time alone, against 86.6 % for temperature. Humidity carries more weather-event content and less calendar content, which is worth remembering when it is used as a regressor.

A caution on the trend panel, and a consequence. NeuralProphet's split between a piecewise-linear trend and a Fourier annual term is not unique, and the two absorb each other: refitting the same series on a different sampling grid moved the inclination's trend standard deviation between 10 and 47 mdeg across runs of this study, with the annual term moving to compensate. The trend panel must therefore not be read as structural drift. It also means the fitted `season_yearly` component is the wrong instrument for measuring seasonal *phase*, which is why Section 5 uses a plain 30-day rolling mean for that purpose instead.

4.2 Imputation of the regressors

Only the regressors are filled, and only with their own trend-plus-seasonal fit — 221 hours each, 2.5 % of the year. The inclinometer keeps its gaps throughout the descriptive analysis: imputing the target and then scoring a forecast against the imputed values would measure the imputation model rather than the forecast.

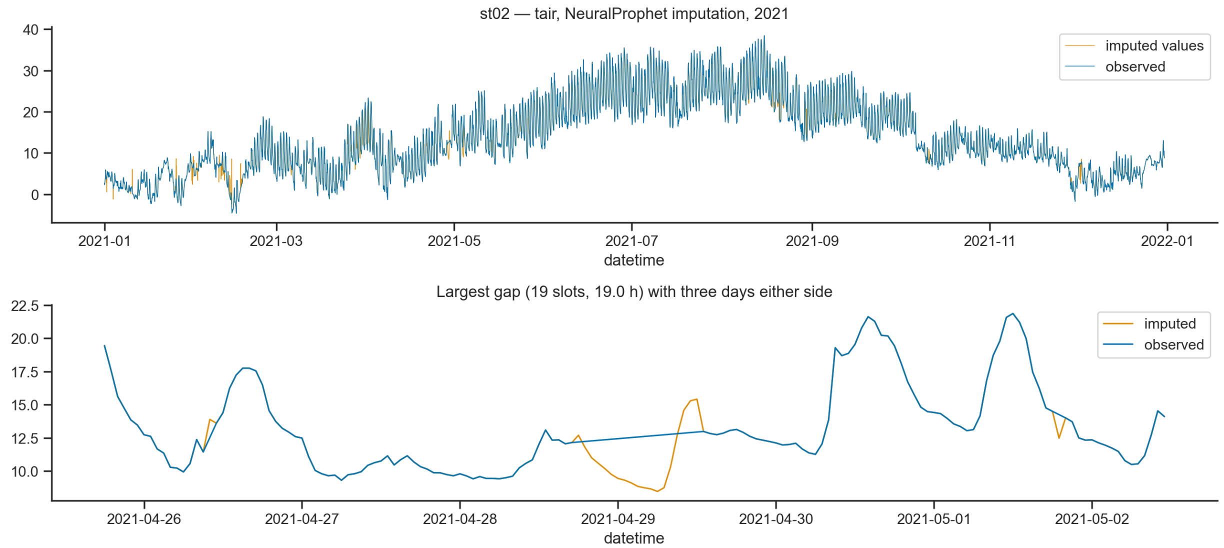

Figure 6: Air temperature: the full year with imputed values overlaid (top), and the largest single gap with three days of context either side (bottom).

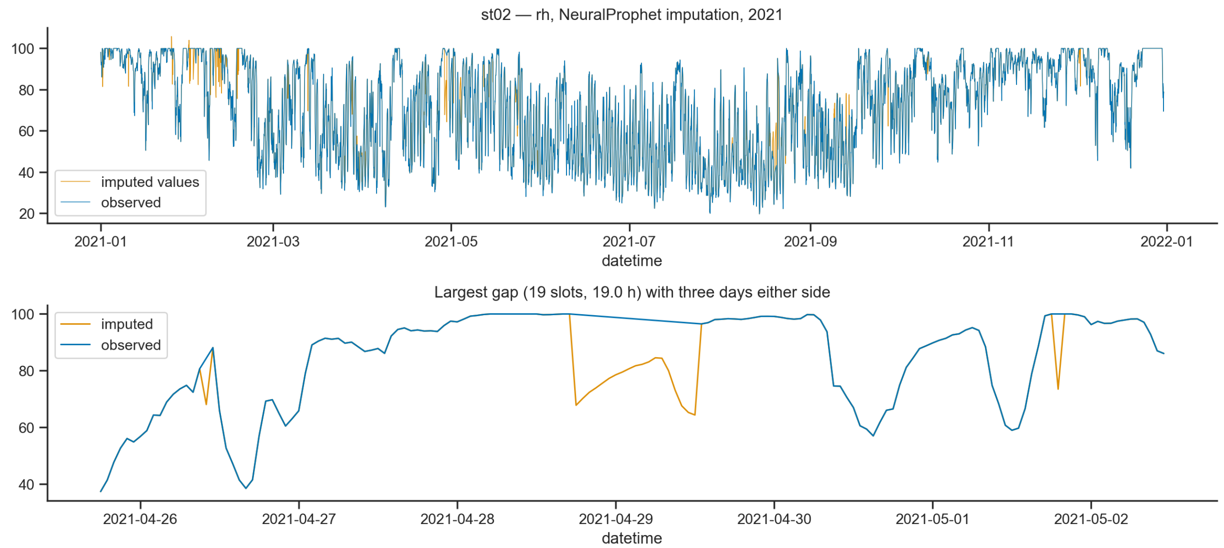

Figure 7: Relative humidity, filled on the same basis.

The detail panel is the honest view of what this imputation can and cannot do. Across the largest gap the fill reproduces the diurnal shape and joins smoothly to the observed data at both ends, which is what a trend-plus-seasonal model should achieve. What it cannot reproduce is that day's particular weather: the filled segment is a smooth climatological day, while the surrounding observed days carry sharp weather-driven excursions of several degrees. Filled values are therefore systematically less variable than observed ones.

5 Is the response instantaneous?

This is the section the full year makes possible, and its answer differs by timescale.

5.1 Correlation structure: levels against differences

Table 6: Pearson correlations over 2021, on the hourly grid.

Pair	Levels	First differences
`inc` − `tair`	+0.238	+0.866
`inc` − `rh`	−0.405	−0.696
`tair` − `rh`	−0.692	−0.706

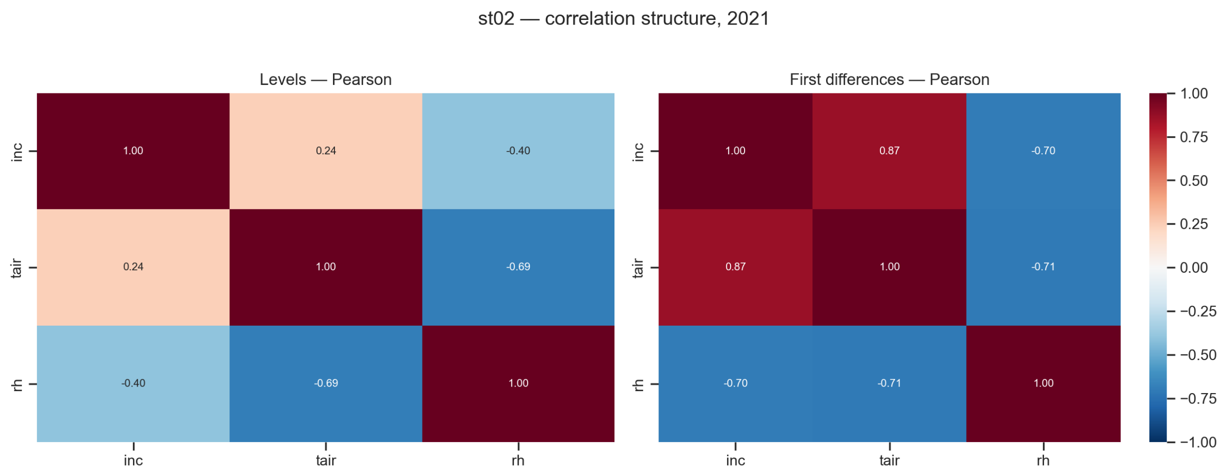

Figure 8: Correlation structure over 2021.

Ordinarily a correlation in levels between two trending series is *inflated* by the shared trend and falls under differencing. The inclination–temperature pair does the opposite, and by a wide margin: 0.238 in levels against 0.866 in differences. A relationship that is weak in levels and strong in differences is the signature of two series that move together at short timescales while their slow components are out of step.

5.2 Lag structure at the daily scale

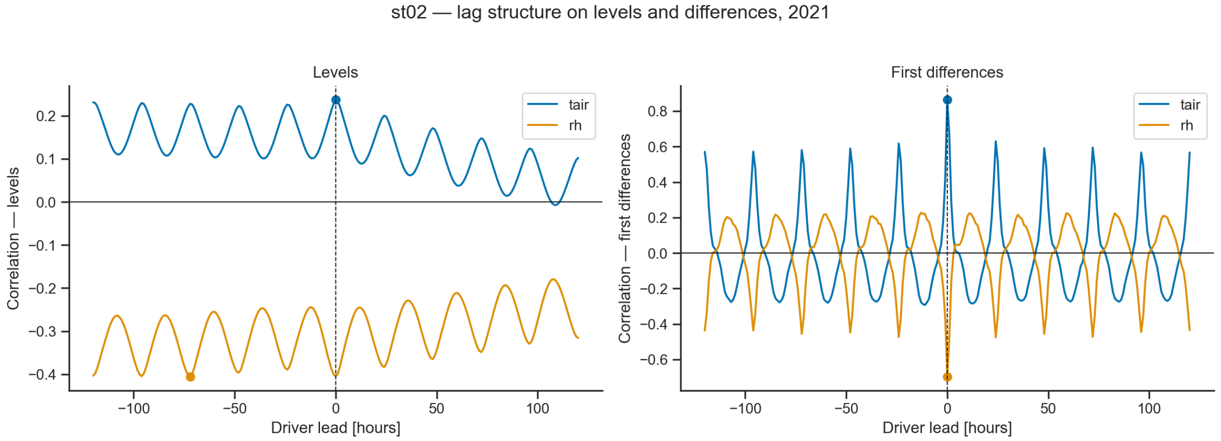

Figure 9: Cross-correlation of the inclination against each driver over ±120 hours, on levels (left) and on first differences (right).

Table 7: Peak of each cross-correlation function at the daily scale.

	Levels		First differences	
Driver	**Peak** r	**Lag**	**Peak** r	**Lag**
`tair`	+0.238	0 h	+0.866	0 h
`rh`	−0.405	0 h	−0.696	0 h

Both drivers peak exactly at zero lag on both views. At the daily scale the response is contemporaneous, and the instantaneous form of Equation (1) is not the problem — the same conclusion the current-era study reached from richer data.

5.3 Thermal inertia

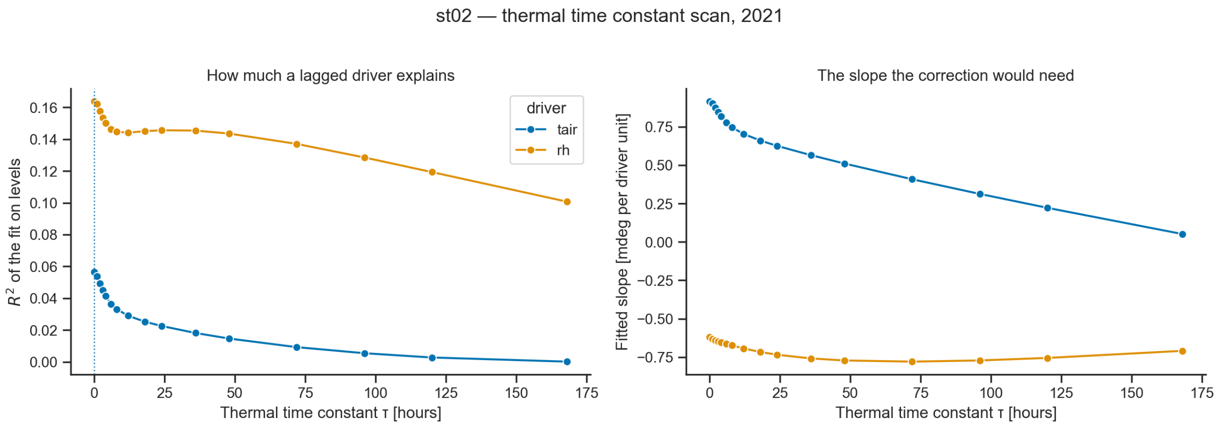

Figure 10: Explained variance and fitted slope against thermal time constant, full signal, 2021.

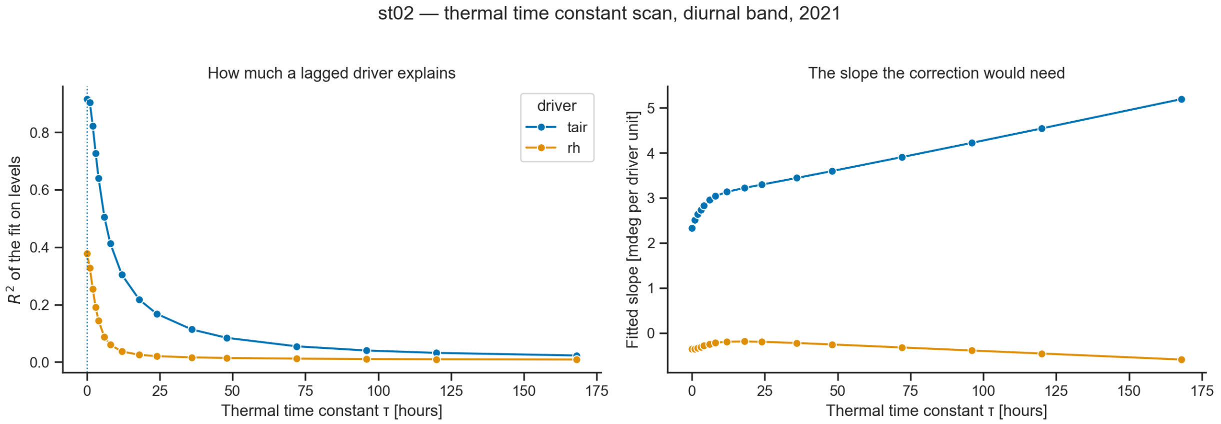

Figure 11: The same scan after removing a centred weekly mean, which restricts the comparison to the diurnal band.

Table 8: Best time constant per driver. Every scan returns zero, and every non-zero time constant makes the fit worse.

	Full signal			Diurnal band		
Driver	**Best** τ	R^2**,** τ**=0**	R^2**, best**	**Best** τ	R^2**,** τ**=0**	R^2**, best**
`tair`	0 h	0.056	0.056	0 h	**0.915**	**0.915**
`rh`	0 h	0.164	0.164	0 h	0.378	0.378

The contrast between the two halves of this table is the study's central measurement. At the **diurnal** scale, instantaneous air temperature explains 91.5 % of the variance of the inclination — a very tight, undelayed relationship, matching the 93.2 % measured in the current era on a different instrument. At the **annual** scale it explains 5.6 %. Adding a thermal time constant does not repair the annual fit; it degrades it monotonically, from 0.056 at $\tau = 0$ to 0.000 at $\tau = 168$ hours.

5.4 The seasonal scale: the structure leads the air

A lag scan reaching five days cannot address an annual relationship, and the obvious extension has a trap in it: the cross-correlation of two near-annual cycles is itself near-annual, so the largest *absolute* correlation sits about half a year away where the two are simply in anti-phase. That is a property of periodicity, not a lag. Two estimators are therefore used, both immune to it.

Table 9: Seasonal phase, measured two ways. The rolling-mean estimator assumes nothing about the shape of the cycle; the cross-correlation is restricted to lags carrying the sign of the contemporaneous correlation.

Series	**Peak of 30-day mean**	**Seasonal range**	**Lead over `tair`**
`inc`	7 June 2021	90.0 mdeg	**59 days**
`tair`	5 August 2021	24.8 °C	—
`rh`	11 January 2021	47.6 %	206 days
Constrained cross-correlation of daily means, ±150 *days*			
`inc` vs `tair`	peak $r = +0.798$	at −89 days	$r = +0.164$ at lag 0

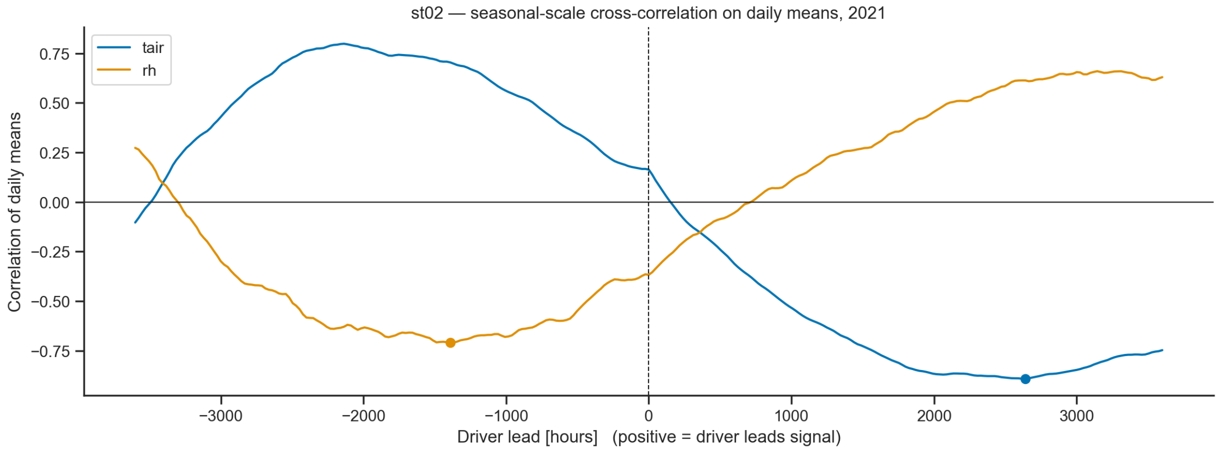

Figure 12: Cross-correlation of daily means over ±150 days. The near-annual oscillation of the curve itself is the periodicity artefact described above; the physically meaningful feature is that the positive peak sits at a negative lag.

Both estimators agree on the sign and on the order of magnitude: **the inclination reaches its seasonal maximum roughly two to three months before air temperature does**. Three consequences follow, and they are the reason this study exists.

No thermal lag can produce this. A causal filter delays; it never advances. That is why the inertia scan finds $\tau = 0$ and why every larger value is worse: the model family being searched cannot represent a lead, so the best it can do is not to move.

The annual movement is not a response to air temperature. Whatever drives it arrives before the air does. Solar gain on an exposed face is the natural candidate, because irradiance on a vertical surface peaks well before the solstice, when the sun is lower and strikes the wall more squarely — and the geometry of an embankment, with one face exposed and one restrained, is exactly the configuration that converts that gain into a tilt. This study cannot confirm it: the legacy era has no radiometer. It can only establish that air temperature is the wrong driver at this timescale, and say what to measure next.

One instantaneous coefficient cannot serve both timescales. The daily response is tight, contemporaneous and worth $R^2 = 0.92$; the annual response is loose and phase-shifted. Equation (1) applies a single constant to both.

6 How large is the thermal response over a full year?

Table 10: Univariate regression of the inclination on each driver over 2021, with Newey–West standard errors. The documented coefficient implies 5.0 mdeg/°C.

Driver	Slope	95 % CI	SE	R^2	n
Levels					
`tair`	0.9139	0.6287 – 1.1991	0.1455	0.057	8 516
`rh`	−0.6201	−0.7355 – −0.5046	0.0589	0.164	8 516
First differences					
`tair`	2.2603	2.1804 – 2.3402	0.0408	0.750	8 432
`rh`	−0.4729	−0.5003 – −0.4455	0.0140	0.484	8 432

The two timescales give 0.91 and 2.26 mdeg/°C — a factor of 2.5 between them, and neither near 5.0. The R^2 values tell the same story as Section 5: instantaneous air temperature is a poor explanation of *where* the inclination sits over a year (0.057) and an excellent explanation of *how it is moving* from hour to hour (0.750).

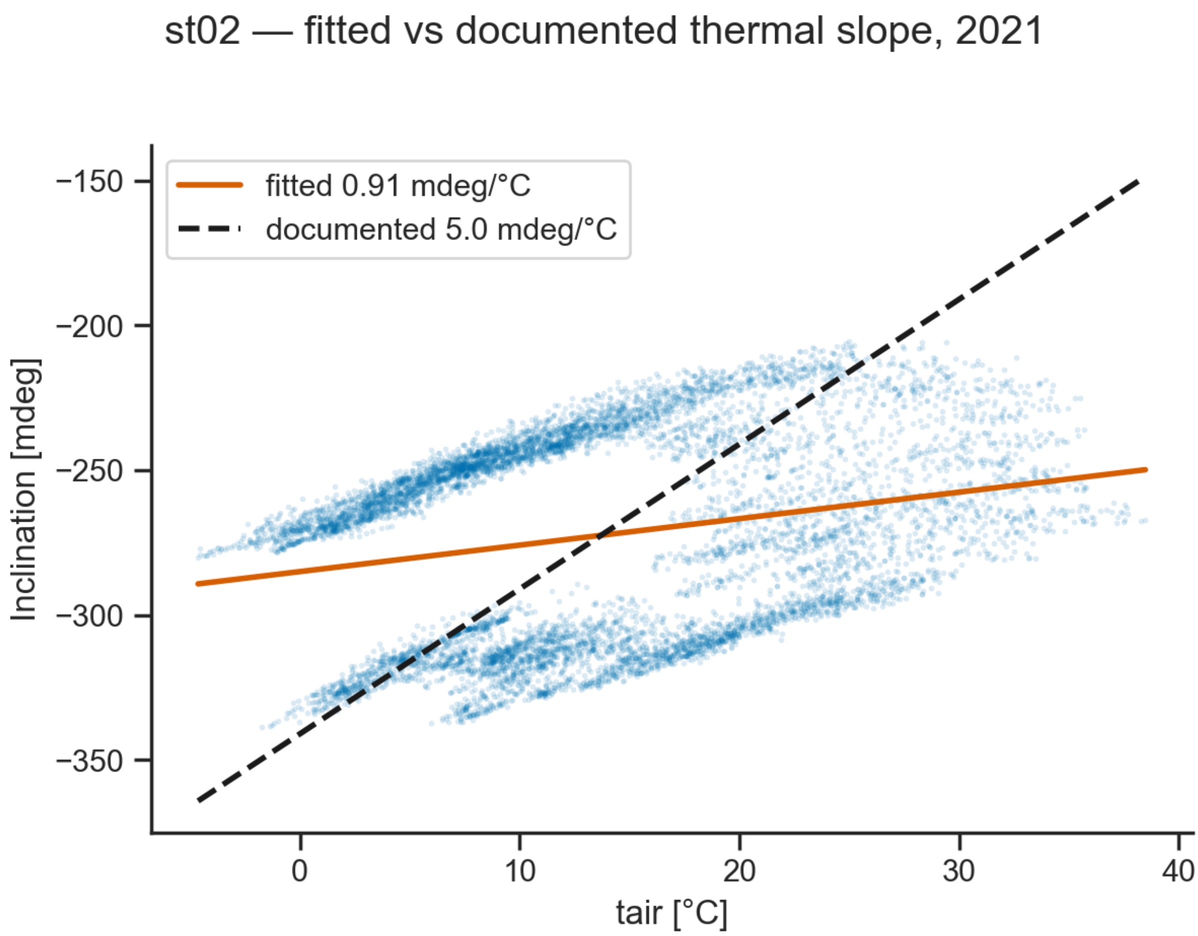

Figure 13: Inclination against air temperature over 2021, with the fitted slope and the slope implied by the documented coefficient.

Figure 13 is what a phase-shifted annual cycle looks like in a scatter plot: not a line with scatter about it but a broad loop. The same air temperature corresponds to inclinations tens of millidegrees apart depending on whether the year is warming or cooling at that moment. The relationship is genuinely multi-valued in levels, so no instantaneous scalar correction, at any coefficient, can collapse it.

7 Acceptance tests

Table 11: Acceptance tests over 2021. The target for every row is a *lower* value after correction. Six of seven fail.

Test	Raw	Compensated	Change	Verdict
Variance [mdeg^2]	1148.58	2381.76	+107.4 %	worse
Standard deviation [mdeg]	33.891	48.803	+44.0 %	worse
Mean diurnal amplitude [mdeg]	21.039	23.920	+13.7 %	worse
$\lvert r\rvert$ with `tair`, levels	0.2378	0.7383	+210.5 %	worse
$\lvert r\rvert$ with `tair`, differenced	0.8657	0.9026	+4.3 %	worse
$\lvert r\rvert$ with `rh`, levels	0.4045	0.3419	−15.5 %	better
$\lvert r\rvert$ with `rh`, differenced	0.6960	0.6972	+0.2 %	worse

Table 12: Signed correlations before and after correction. Both drivers reverse sign.

Driver	Raw	Compensated	Sign reversed
`tair`	+0.2378	−0.7383	yes
`rh`	−0.4045	+0.3419	yes

The single apparent improvement is again an artefact of passing through zero on the way to a large correlation of the opposite sign. The correction more than doubles the variance of the series it is supposed to clean.

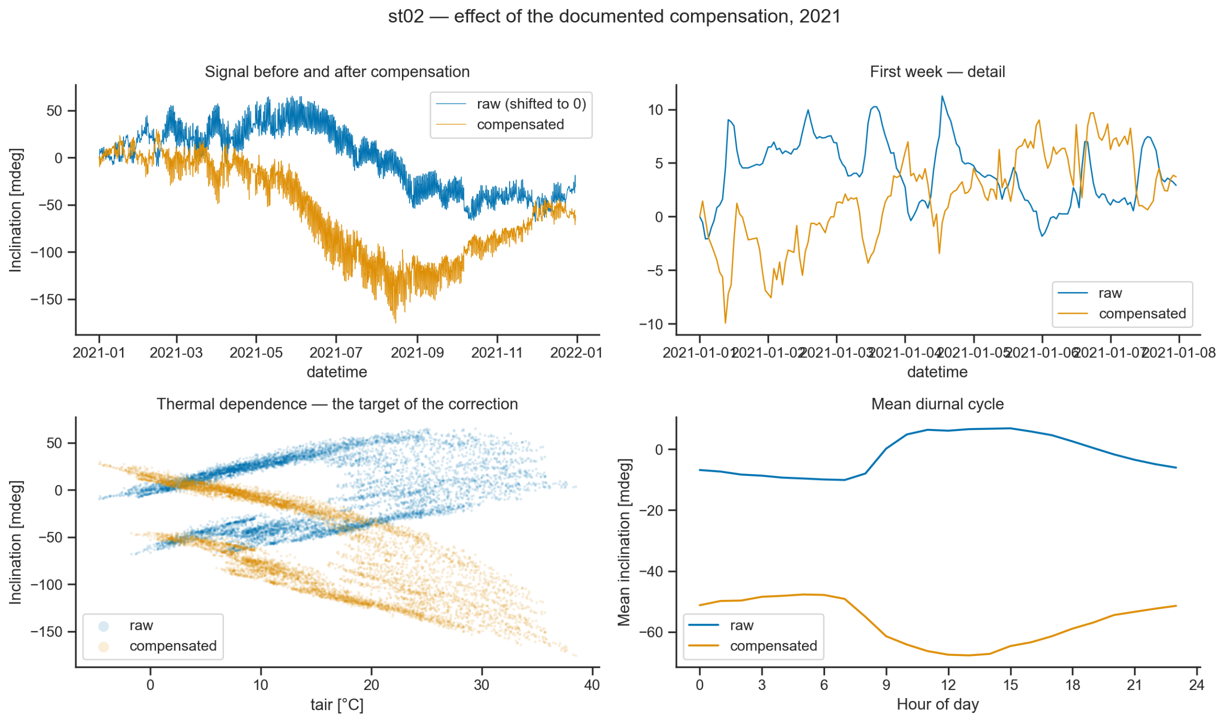

Figure 14: Effect of the documented correction over 2021. Top left, the full year before and after; top right, the first week in detail; bottom left, the thermal dependence; bottom right, the mean diurnal cycle.

Full year. The raw series spans roughly 110 mdeg peak to trough. The compensated series spans more than 200, plunging to about −170 mdeg at the end of summer before recovering. The correction did not remove a seasonal excursion; it roughly doubled one and inverted its phase.

First week. In January the raw and compensated traces are already near mirror images at daily scale.

Thermal dependence. The raw cloud is the broad loop of Figure 13. The compensated cloud is a tighter, steeply descending band — a stronger dependence on temperature than before, not a weaker one.

Mean diurnal cycle. The raw profile rises from about −10 mdeg overnight to +7 mdeg at midday. The compensated profile is inverted and larger.

8 Where the documented coefficient sits

Table 13: Selected points from the 2021 coefficient sweep.

	Coefficient [mdeg/°C]	Residual variance [mdeg^2]	$\lvert r\rvert$ with `tair`
	−1.0	1368.45	0.456
No compensation	0.0	1148.58	0.238
	0.8	1084.65	0.031
Variance-minimising	**1.0**	**1084.22**	**0.023**
	1.2	1090.00	0.076
	2.0	1175.35	0.279
	3.0	1421.99	0.488
Documented	**5.0**	**2381.76**	**0.738**
	7.0	3963.53	0.852

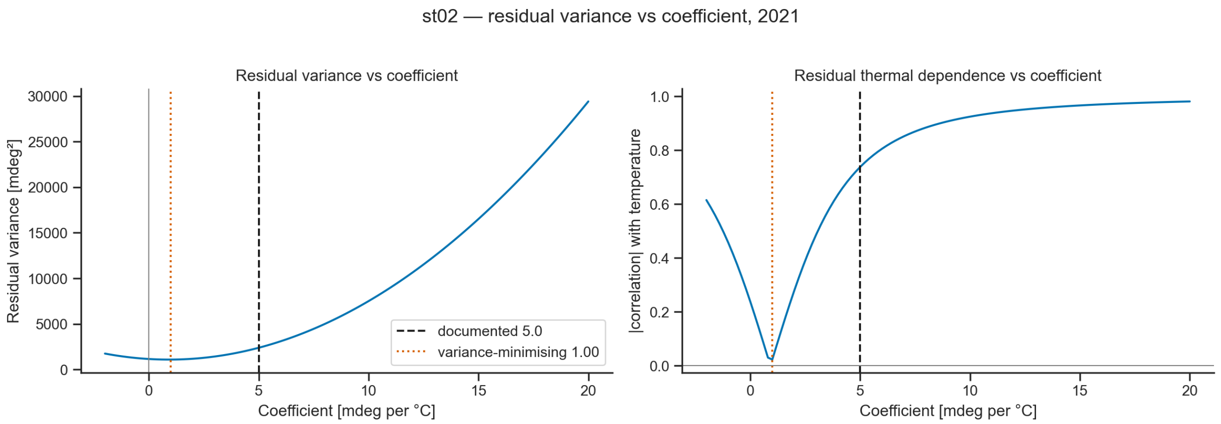

Figure 15: Residual variance and residual thermal correlation against the compensation coefficient over 2021, with the documented value marked.

Two readings, the second specific to the full-year setting.

First, the documented coefficient is five times the variance-minimising one and sits far up the ascending branch. Applying it more than doubles the residual variance relative to leaving the signal alone — 2382 against 1149 mdeg^2.

Second, **even the optimal coefficient buys very little here**. The minimum, 1084 mdeg^2, is under six per cent below the do-nothing value. In the current era the same sweep cut residual variance by seventy per cent. The reason is Section 5: over a full year the dominant thermal component is out of phase with instantaneous temperature, and an instantaneous linear term cannot remove what is not aligned with it. The residual thermal correlation still falls almost to zero at the optimum (0.023), so the correction can null the *correlation* while barely touching the *variance* — a useful reminder that those are different targets, and that a compensation tuned on one may be worth almost nothing on the other.

9 The prediction experiment

The configurations, the raw-scale scoring rule and the ridge reference model are those of the companion study, with `tair` and `rh` as the only available regressors. C against D is the controlled contrast. Predictions of a compensated target have the correction added back before the error is computed, so all configurations are scored on the same physical quantity.

E and F degenerate here, and that is a result. Configuration E shifts each driver to its measured best lag and F filters each through its measured best time constant. In this era both measurements returned zero for both drivers, so E and F are *identical to A* by construction and score identically. Reporting them is the point: the physics-motivated configurations were given every chance to differ and the data declined to make them differ, at the timescale a forecasting model operates on.

Table 14: 2021, one-hour-ahead ridge forecast, 5 436 training and 2 330 test samples.

Configuration	MAE [mdeg]	RMSE [mdeg]	vs best
A, E, F · raw + regressors	**0.9878**	1.5088	**best**
D · raw, no regressors	1.0480	1.5494	+6.1 %
C · compensated, no regressors	1.5443	2.2041	+56.3 %
B · compensated + regressors	2.0760	2.6780	+110.2 %

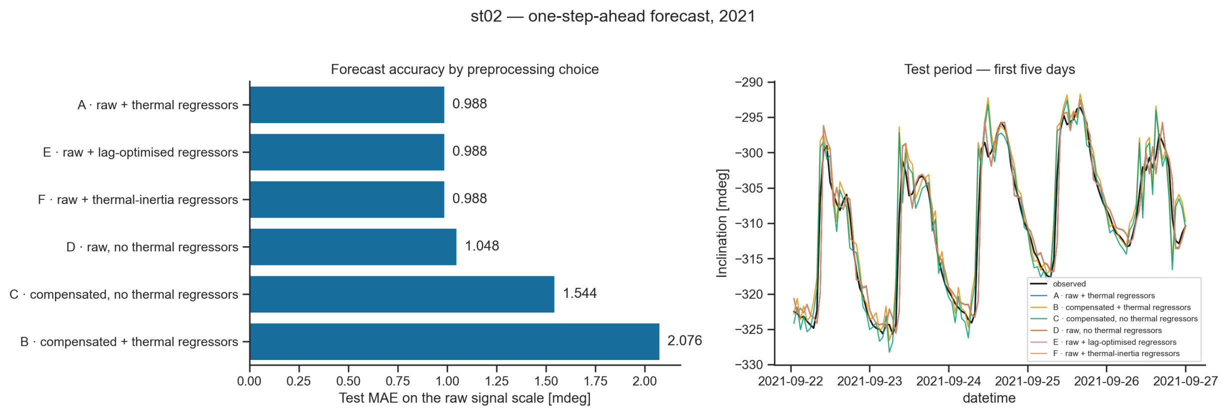

Figure 16: One-hour-ahead forecast, ridge model, 2021.

Table 15: 2021, 24-hour-ahead ridge forecast, 5 350 training and 2 294 test samples.

Configuration	MAE [mdeg]	RMSE [mdeg]	vs best
A, E, F · raw + regressors	**3.9242**	5.1083	**best**
D · raw, no regressors	4.0404	5.2348	+3.0 %
C · compensated, no regressors	4.0692	5.2063	+3.7 %
B · compensated + regressors	7.9082	8.8802	+101.5 %

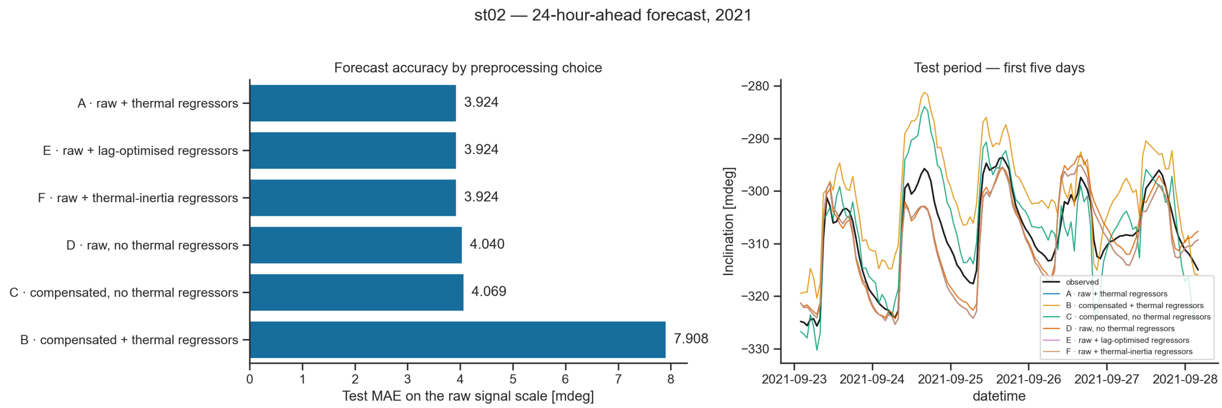

Figure 17: Twenty-four-hour-ahead forecast, ridge model, 2021.

Unlike the current-era study, supplying the drivers helps here — A beats D at both horizons. That is what a full annual cycle buys: with a whole year in the training split the model is not forced to extrapolate outside its training temperature range, and the confound that made the current-era A-versus-D comparison uninformative is absent.

Table 16: The controlled contrast, C against D.

Horizon	MAE raw (D)	MAE compensated (C)	Penalty
1 h	1.0480	1.5443	+47.4 %
24 h	4.0404	4.0692	+0.7 %

The penalty is large at one hour and almost nil at twenty-four. The reason is the same one that made the coefficient sweep so flat: at the daily horizon the error is dominated by the part of the signal that an instantaneous thermal term cannot describe either way, so compensating the target changes little. It still changes it for the worse, and configuration B — compensated *and* given the regressors — remains catastrophic at +101.5 %, the double-correction signature seen in the current era.

9.1 Confirmation under NeuralProphet

Table 17: 2021, one hour ahead under NeuralProphet, 2 600 scored points per configuration, fixed random seed.

Configuration	MAE [mdeg]	RMSE [mdeg]	vs best
A, E, F · raw + regressors	**2.7765**	3.5824	**best**
D · raw, no regressors	3.0151	3.7775	+8.6 %
C · compensated, no regressors	3.2906	4.0972	+18.5 %
B · compensated + regressors	3.4718	4.3762	+25.0 %

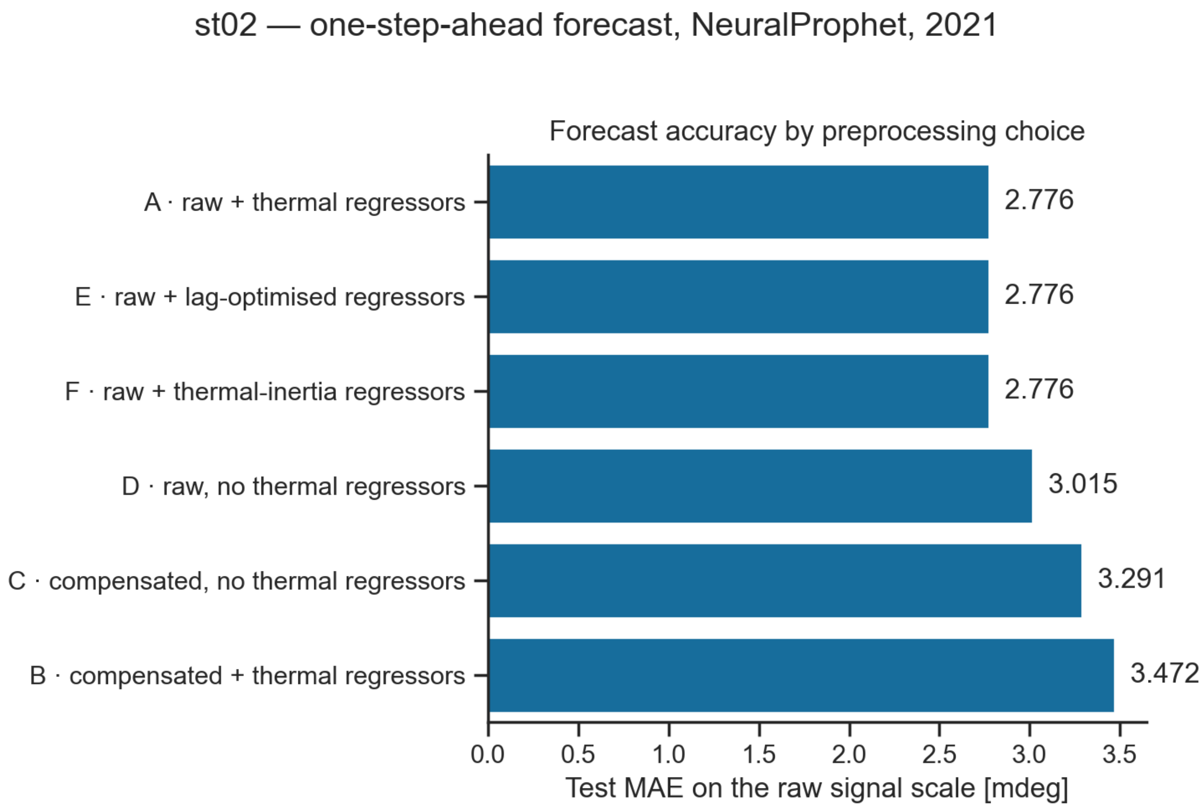

Figure 18: One-hour-ahead forecast under NeuralProphet, 2021.

Under the deployed model the ordering is the same as in the current era: both raw configurations beat both compensated ones. The margins are smaller than the ridge model's, as they should be — NeuralProphet's own seasonal terms absorb part of what the correction damages.

10 Verdict

Three criteria, three failures — on independent data

1. Thermal dependence is not reduced. $|r|$ with air temperature rises from 0.238 to 0.738 and reverses sign.

2. The coefficient is not near the variance-minimising value. 5.00 against 1.00 mdeg/°C, more than doubling residual variance relative to doing nothing.

3. The forecast does not improve. The controlled-contrast penalty is +47.4 % at one hour and +0.7 % at twenty-four, and +18.5 % under NeuralProphet. No horizon and no model favoured the compensated target.

10.1 Across both eras

Table 18: The same quantities, measured independently in each era.

Quantity	Legacy 2021	Current era	Documented
Fitted `tair` slope, levels [mdeg/°C]	0.91	1.63	**5.0**
Fitted `tair` slope, differences [mdeg/°C]	2.26	2.17	**5.0**
Variance-minimising coefficient [mdeg/°C]	1.00	1.60	**5.0**
Diurnal-band R^2, instantaneous `tair`	0.915	0.932	—

Two things are worth noting in this table. The differenced slopes agree closely across eras and instruments — 2.26 against 2.17 mdeg/°C — as do the diurnal-band R^2 values, 0.915 against 0.932. **The fast thermal response of this wall is a stable, reproducible property.** The levels slopes do not agree, because they are dominated by the seasonal behaviour, which differs between a half cycle and a full one.

Together with the three legacy stations measured from the supplied `Data.mat` — 7.21, 1.33 and 9.66 mdeg/°C — there are now five independent estimates of the levels slope spanning 0.91 to 9.66. **No single constant serves them, and none is near** 5.0**.**

10.2 Recommendation

Remove the fixed compensation from Notebook 00. The recommendation is unchanged from the current-era study and now rests on two independent datasets, two instruments, two regressor sets and two model families.

Beyond that, this study says how the thermal response should be handled instead, and the answer is that the two timescales must be separated:

- **The daily response** is contemporaneous, tight and reproducible ($R^2 = 0.92$ on the diurnal band, instantaneously). Instantaneous air temperature is an excellent regressor for it, and no lag or time constant improves on that.
- **The seasonal response** leads air temperature by roughly two months and cannot be represented by any coefficient or lag on it. It belongs in the decomposition's own annual term, or in a driver that carries the right phase. Installing a radiometer at the legacy stations, or reconstructing irradiance on the wall's actual orientation from a reanalysis product, is the measurement that would settle it.

11 Caveats

- One calendar year gives exactly one pass through the annual cycle, so the seasonal lead is measured once and cannot be checked against a second cycle. Read it as a description of 2021. Two estimators agree on its sign and order of magnitude but differ on its size (59 against 89 days), which is the honest precision available.
- The attribution of the seasonal lead to solar geometry is an interpretation, not a measurement. The legacy era has no radiometer; this study establishes only that air temperature is the wrong driver at that timescale.
- The trend and annual components of the decomposition are not separately identifiable, as Section 4.1 shows explicitly. Neither is read as structural drift.
- Only the regressors are imputed (2.5 % of hours) and filled values are systematically smoother than observed ones. The inclinometer keeps its gaps.
- In the NeuralProphet experiment, gaps in the model input are bridged by interpolation so the autoregressive window has continuous history; scoring happens only at observed timestamps, identically for all configurations.
- `lc_lib.py` imports from `../thermal_compensation/tc_lib.py`. Moving or renaming either study folder breaks that import.

A Reproduction

```
conda activate neuralprophet_env
cd studies/thermal_compensation_legacy
jupytext --to ipynb legacy_compensation_study.py
jupyter nbconvert --to notebook --execute --inplace \
        legacy_compensation_study.ipynb
```

Edit the paired `.py` file, never the `.ipynb`. Setting `CONTEXT = 'paper'` in the parameter cell re-exports every figure at manuscript column size. The archive is never written to.

One library workaround is in force. NeuralProphet 0.8.0 raises `UnboundLocalError: conditional_cols` when a prediction frame ends in `NaN` and imputation is disabled — the variable is bound only inside the imputation branch but is read again when trailing rows are restored. The hourly grid ends in a gap where the 20-minute grid did not, so `decompose_series` trims the trailing gap before predicting and reindexes the components back onto the full span, leaving those hours empty.

B Output inventory

Table 19: Artefacts produced by the notebook. Identifiers are stable names, not an ordering.

Identifier	Contents	In this report
S2_01	Year completeness scan	Table 2
S2_02	Year completeness figure	Fig. 1
S2_03	Channel coverage, 2021	Table 4
S2_04	Gap profile	Table 3
S2_05	Year overview	Fig. 2
S2_06	Decomposition, three series	Figs. 3–5
S2_07	Decomposition summary	Table 5
S2_08	Imputation, `tair` and `rh`	Figs. 6, 7
S2_09, S2_10	Correlation matrices	Table 6
S2_11	Correlation heatmaps	Fig. 8
S2_12, S2_31	Cross-correlation peaks, differences and levels	Table 7
S2_13	Lag structure figure	Fig. 9
S2_14, S2_15	Regression slopes	Table 10
S2_16	Fitted versus documented thermal slope	Fig. 13
S2_17, S2_18	Acceptance tests and signed correlations	Tables 11, 12
S2_19	Effect of the documented compensation	Fig. 14
S2_20, S2_21	Coefficient sweep and figure	Table 13, Fig. 15
S2_22, S2_24	Ridge results, 1 h and 24 h	Tables 14, 15
S2_23, S2_25	Experiment figures	Figs. 16, 17
S2_26	Controlled contrast	Table 16
S2_27, S2_28	NeuralProphet results and figure	Table 17, Fig. 18
S2_29	Verdict	Section 9
S2_30	Cross-era comparison	Table 18
S2_32–S2_35	Inertia scans and best time constants	Table 8
S2_36, S2_37	Inertia scan figures	Figs. 10, 11
S2_38	Lag summary	Section 5
S2_39, S2_41	Seasonal lag and phase	Table 9
S2_40	Seasonal cross-correlation figure	Fig. 12